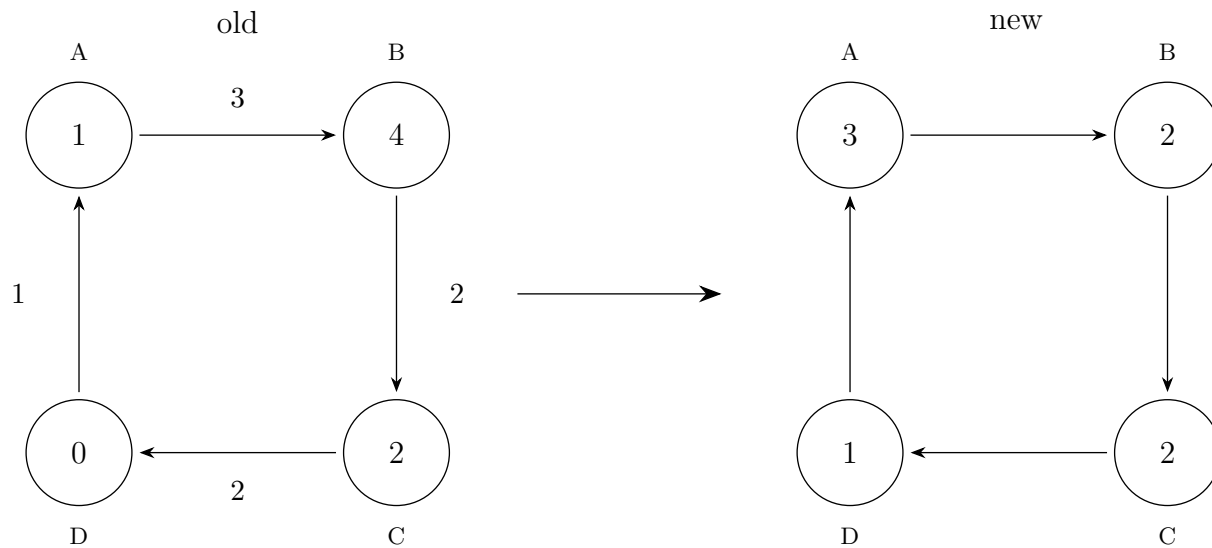
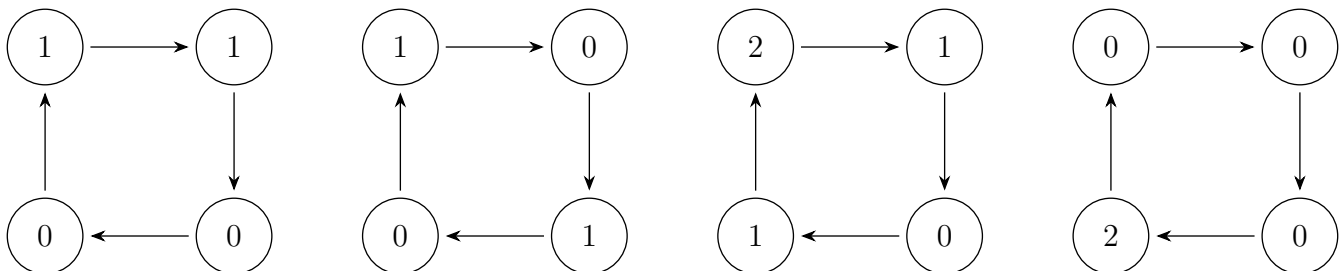


Use whole-number heights, including 0. Follow each arrow: the gap between its two old heights becomes the new height at its tail. Make every new tower before changing any old tower.



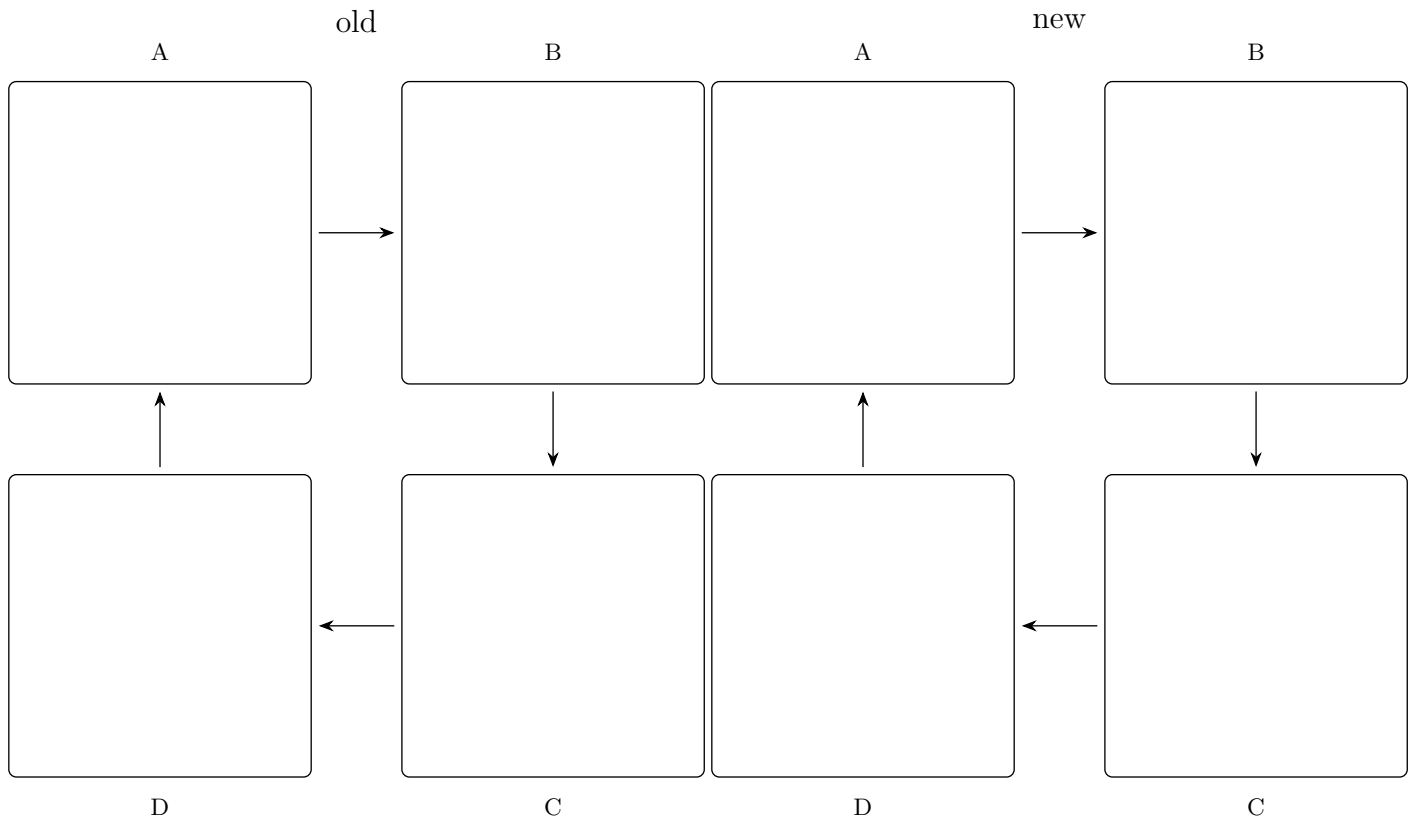
Edge A to B has gap 3, so new A is 3. All four gaps use the old ring.

**Problem 1:** Build these starts and keep making new rings. Stop when all four towers are 0.



**Problem 2:** Choose heights from 0 to 4. Find a start that lasts as many moves as you can; count the move that makes all four towers 0.

An all-zero start takes 0 moves. Use these big mats for all four-station starts.



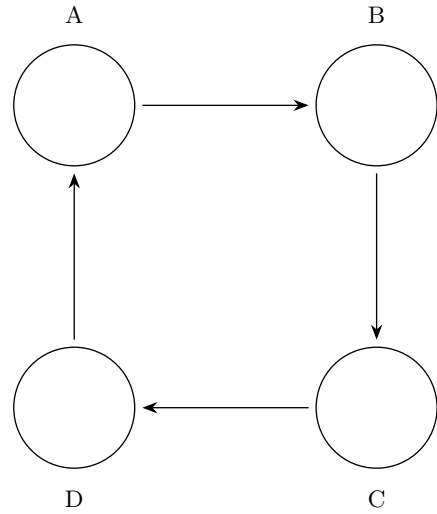
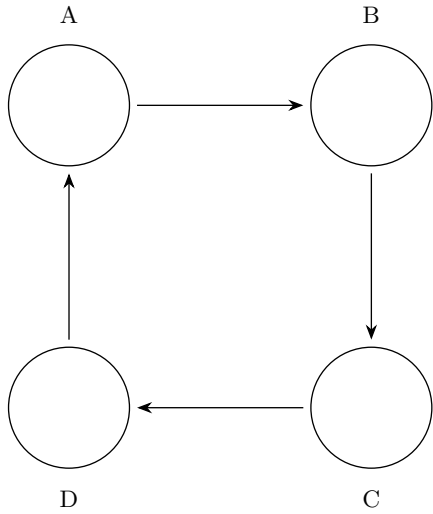
| A | B | C | D |
|---|---|---|---|
|   |   |   |   |

Moves to zero: \_\_\_\_\_

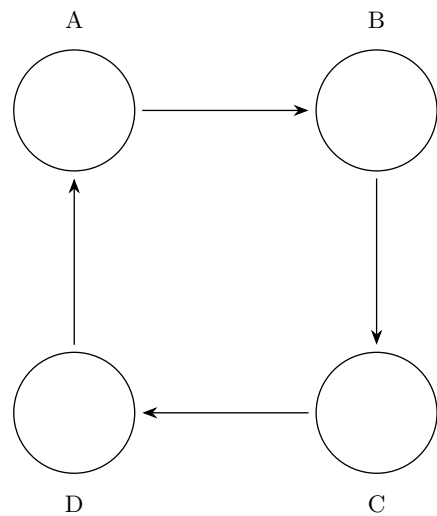
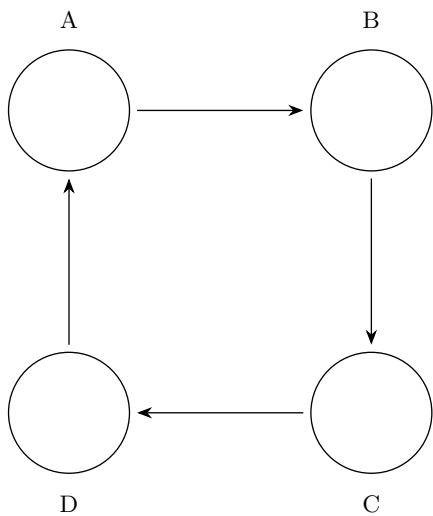
| A | B | C | D |
|---|---|---|---|
|   |   |   |   |

Moves to zero: \_\_\_\_\_

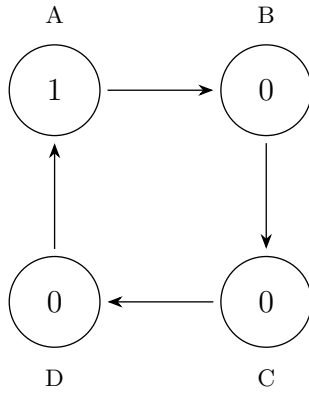
**Problem 3:** Find different starts that become all zeros in one move.



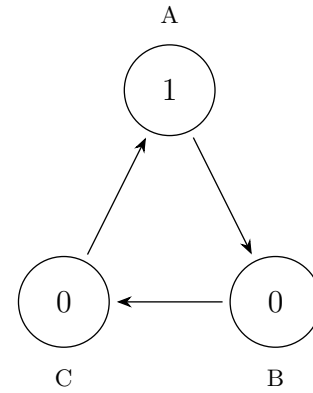
**Problem 4:** Find different starts that first become all zeros on the second move.



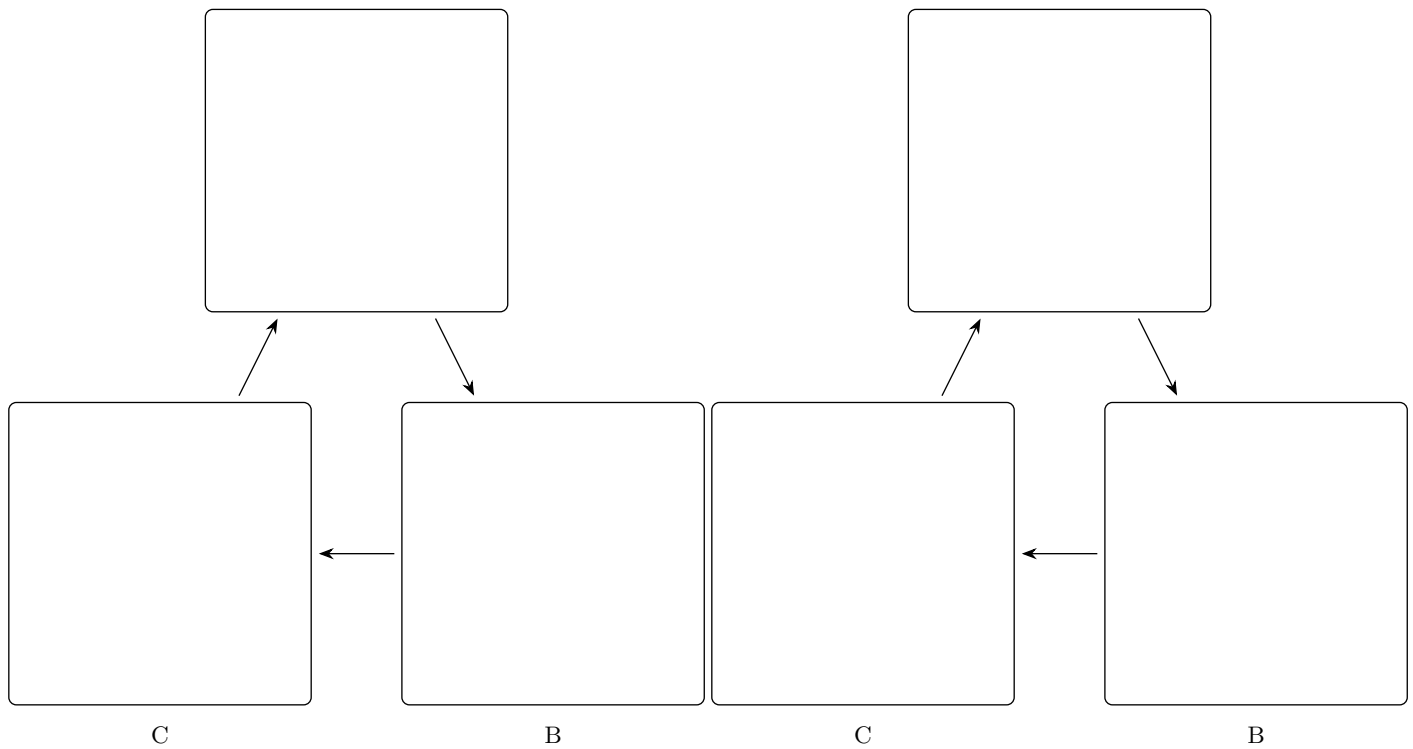
**Problem 5:** Try both starts with the same gap rule. Stop at all zeros or when the whole ring repeats.



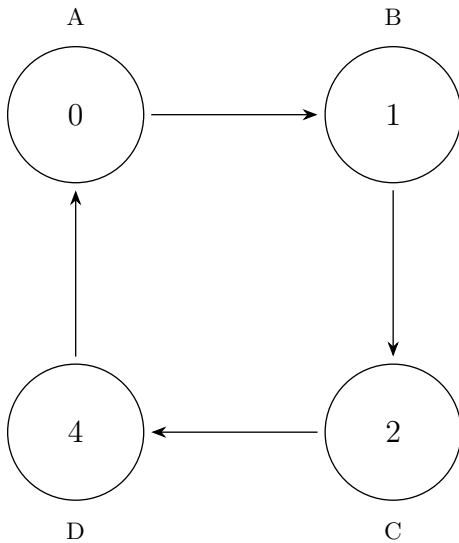
old  
A



new  
A

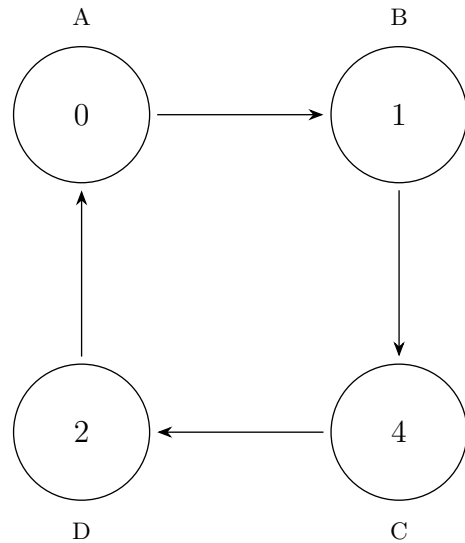


**Problem 6:** Put 0, 1, 2, and 4 around the four stations. Find an order that lasts longer than the other orders you try.



| A | B | C | D |
|---|---|---|---|
|   |   |   |   |

Moves to zero: \_\_\_\_\_



| A | B | C | D |
|---|---|---|---|
|   |   |   |   |

Moves to zero: \_\_\_\_\_

| A | B | C | D |
|---|---|---|---|
|   |   |   |   |

Moves to zero: \_\_\_\_\_